

SIH 2026 PS 26017 — FULL REAL-SOURCE DATA + ML RESEARCH PACK

Prepared for direct handoff to Claude. Web research checked 26 August 2026.

CRITICAL DATA DISTINCTION

REAL: values below are extracted/verified from ADB project pages, indexed ADB completion-report text, or identified government records. **DERIVED:** calculations made from real values. **MISSING:** not publicly verified and intentionally left blank. **SYNTHETIC:** must never be inserted into the real dataset without an explicit label.

The Andri² 2025 paper says its 138-project dataset was collected from ADB and MoSPI and contains project name, country, decision year, project type, estimated cost, actual cost, planned duration, implementation period and causes of delays. The related 2024 paper specifies 108 ADB projects + 30 MoSPI projects. This pack does **not** pretend that the complete 138-row CSV was recovered.

1. REAL PROJECT RECORDS RECOVERED

Project	State	Type	km	Cost \$m	ADB \$m	Counter \$m	Orig close	Rev close	Actual close	Delay mo
013Bihar State Highways II Project	Bihar	State highways	356.0	424.0	300.0	124.7	2014-12-31	—	—	—
National Highway Sector II	Multiple/India	National highways	566.0	670.6	400.0	270.6	2008-12-31	2011-12-31	2012-05-09	4.27
Madhya Pradesh State Roads Secto	Madhya Pradesh	State roads	1800.0	400.0	320.0	80.0	2010-12-31	2011-12-31	2012-10-05	9.17
Karnataka State Highway Improvem	Karnataka	State highways	615.0	452.75	305.0	147.75	2015-06-30	2018-10-31	2019-04-23	5.72
Madhya Pradesh District Connecti	Madhya Pradesh	District roads	1600.0	500.0	350.0	150.0	2018-10-31	—	2019-03-28	—
North Eastern States Roads Inves	Assam/Manipur/Mizoram	State roads	—	157.2	125.2	32.0	—	—	—	—
West Bengal Corridor Development	West Bengal	State highways / rural roads	2260.0	—	148.7	—	—	—	—	—
Rajasthan State Highway Investme	Rajasthan	State highways / MDR	1000.0	—	—	—	—	—	—	—

2. IMPORTANT PROJECT-SPECIFIC EVIDENCE

Bihar State Highways II Project and Additional Financing: ADB PCR recommendation: about 50% of land should be acquired by loan negotiations; contracts should be awarded only after substantial acquisition. Source: <https://www.adb.org/sites/default/files/project-documents/41629/41629-013-44425-013-pcr-en.pdf>

Madhya Pradesh State Roads Sector Project II: ADB project page lists involuntary resettlement as not applicable. Source: <https://www.adb.org/projects/37328-013/main>

Karnataka State Highway Improvement Project: Safeguard category: involuntary resettlement A. Source: <https://www.adb.org/projects/42513-013/main>

Madhya Pradesh District Connectivity Sector Project: Safeguard category: involuntary resettlement B. All 18 civil works contracts were awarded in 2014–2015. Source: <https://www.adb.org/projects/47270-001/main>

Bihar State Highways II: the ADB completion report explicitly recommends about 50% of land be acquired by loan-negotiations stage and substantial acquisition before contract award. The original project documentation also budgets \$78.93m for land acquisition, resettlement/rehabilitation and utility shifting. This is one of the most useful records for land-acquisition feature engineering.

3. DATA SOURCES AND WHAT TO EXTRACT NEXT

ADB PCRs: 108 ADB completion reports underpin the 138-project research dataset according to the 2024 Andriani paper. Each PCR can provide cost, schedule, implementation, and qualitative causes. The complete 108-report corpus still needs systematic collection.

MoSPI/PAIMANA: Indian project monitoring source; use for project-level schedule/cost slippage and reported delay causes.

ADB RRP/PDS: Useful for original planned cost, planned duration, scope, financing and safeguard categories when PCR extraction is incomplete.

Resettlement Plans / Social Monitoring Reports: Best source for parcel/family/compensation/R&R; variables; these are not always in PCRs.

Devi & Sindhu: Use for factor selection and RII/PCA methodology, not as a substitute for project-level training labels.

4. ML DATASET DESIGN

Project variables: cost, project type, state, planned duration, road length

Land acquisition: land required, land acquired, acquisition %, affected parcels, affected families

Compensation: required amount, paid amount, compensation %, pending cases

Legal: pending legal cases, disputes, injunctions

R&R: R&R; completion %, affected households, resettlement progress

Clearances/ROW: forest/environment clearance, encroachment, utility shifting, ROW/ROU issues

Administrative: approval delay, agency count, procurement delay

Target: time overrun months or %, and/or delayed/not delayed

5. FORMULAS

- Time-overrun % = $100 \times (\text{Actual duration} - \text{Planned duration}) / \text{Planned duration}$
- Cost-overrun % = $100 \times (\text{Actual cost} - \text{Estimated cost}) / \text{Estimated cost}$
- Acquisition % = $100 \times \text{Land acquired} / \text{Land required}$
- Compensation paid % = $100 \times \text{Compensation paid} / \text{Compensation required}$
- Linear regression: $y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \varepsilon$
- Logistic regression: $P(\text{delay}) = 1/(1 + e^{-(z)})$, $z = \beta_0 + \sum \beta_i X_i$
- RII = $\sum W/(A \times N)$, with $A=5$ for a 5-point survey
- Legal intensity = $\text{pending legal cases}/(\text{affected parcels} + 1)$

6. MODEL PLAN

Baseline: linear regression + logistic regression. Main model: Random Forest or XGBoost. Explainability: SHAP/permutation importance. Regression metrics: MAE, RMSE, R^2 . Classification metrics: precision, recall, F1, ROC-AUC. Use cross-validation and a held-out test set. Explicitly check target leakage. For early detection, do not use information that only becomes known after the outcome.

7. SIH PRODUCT OUTPUT

Dashboard should return: delay probability, predicted months of delay, risk band, top contributing factors, affected acquisition stage, and recommended intervention. Example output is a design illustration only: 84% risk, 9.8 predicted months, with acquisition completion, legal cases and compensation completion as leading contributors.

8. REFERENCES / OFFICIAL SOURCE LINKS

Andriani et al. 2025 — infrastructure delays
<https://www.sciencedirect.com/science/article/pii/S1226798825003241>

Andriani et al. 2024 — cost overruns
<https://www.mdpi.com/2071-1050/16/24/11159>

ADB — Bihar State Highways II PCR
<https://www.adb.org/projects/documents/ind-41629-013-44425-013-pcr>

ADB — National Highway Sector II
<https://www.adb.org/projects/35335-013/main>

ADB — Madhya Pradesh State Roads II
<https://www.adb.org/projects/37328-013/main>

ADB — Karnataka State Highway Improvement
<https://www.adb.org/projects/42513-013/main>

ADB — Madhya Pradesh District Connectivity
<https://www.adb.org/projects/47270-001/main>

ADB — North Eastern States Roads Project 2
<https://www.adb.org/projects/37143-033/main>

ADB — West Bengal Corridor Development
<https://www.adb.org/projects/32203-013/main>

ADB — Rajasthan State Highway Investment Tranche 1
<https://www.adb.org/projects/49228-002/main>

MoSPI PAIMANA
<https://paimana-proj.mospi.gov.in/>

MoSPI public dashboard
<https://ipm.mospi.gov.in/Home/PublicDashboard>

Devi & Sindhu 2025
<https://link.springer.com/article/10.1007/s40030-025-00899-5>

9. HANDOFF INSTRUCTION TO CLAUDE

Use the accompanying XLSX as the structured data workbook and the CSV as the machine-readable real-source table. First expand the ADB corpus systematically; then extract PCR fields into the same schema. Never fill unknown fields by guessing. Keep source_url and source_page/source_section for every extracted value. Build a separate synthetic-demo table only for missing parcel-level land-acquisition variables.